Ruhila Goswami

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RuhiRG

Applicant Details

Current name Ruhila Goswami
Previous name Ruhila S.

Research Profile

FOCUS Evolutionary divergence in salmonids, linking dental and craniofacial phenotypes with population history, genome structure, and selection.

INTERESTS Population genetics; phenotypic-genotypic mapping; structural variation in genomes and selection; salmonid evolution; non-invasive methods.

METHODS Phylogenetic inference, genomic synteny, PCR-based sex determination, skeletal phenotyping, field sampling, and reproducible analysis in R and Python.

Education

2024–2026 **M.S. Biological Science**, *University of Iceland*, Reykjavík, *Iceland*
(ADVISOR: Prof. Arnar Pálsson)

THESIS: Evolutionary Divergence in Dental Variation and Sex Chromosome Architecture in Icelandic Brown Trout and Arctic Charr

2021–2024 **B.S. Biology (Major)**, *Indian Institute of Science Education and Research (IISER)*, Mohali, *India*
First Division with Honors

2020 **Intermediate (AISSCE)**, *Velammal Vidyalaya, Ayanabakkam*, Chennai, *India*
83.8% Central Board of Secondary Education (CBSE)

2017 **High School (AISSE)**, *Velammal Vidyalaya, Ayanabakkam*, Chennai, *India*
10.0 Cumulative Grade Point Average (CGPA) in Central Board of Secondary Education (CBSE)

Publications

Manuscripts in Preparation

In preparation — Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids

Conference Proceedings

Rohit Goswami, Ruhila S , Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. "Reproducible High Performance Computing Without Redundancy with Nix." In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

Rohit Goswami and Ruhila S . "High Throughput Reproducible Literate Phylogenetic Analysis (accepted)." In: *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*. 2023.

Preprints

Rohit Goswami, Ruhila S , Amrita Goswami, Sonaly Goswami, and Debabrata Goswami. *Unified Software Design Patterns for Simulated Annealing*. Feb. 6, 2023. arXiv: `2302.02811 [physics]`. URL: `http://arxiv.org/abs/2302.02811` (visited on 02/10/2023).

Presentations and Posters

Posters

April 2026 — **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *Icelandic Ecological Society (Vistís)*, Ruhila Goswami

March 2026 — **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *NOWPAS*, Ruhila Goswami

October 2025 — **Evolutionary divergence in tooth numbers and bone shape by populations and species in Icelandic salmonids**, *IceBio*, Ruhila Goswami

2025 — **Diversity in craniofacial elements by populations and sexes in salmonids of Iceland**, *Icelandic Ecological Society (Vistís)*, Ruhila Goswami, Best Poster Award

November 2022 — **Tracing Lineages of *Salmo salar* through Histone sequence data**, *BES Annual Meeting 2022*, Ruhila S. Goswami (née S.)

Oral Presentations

November 2022 — **High Throughput Reproducible Literate Phylogenetic Analysis**, *IEEE PDGC 2022*, R. Goswami, Ruhila S. Goswami (née S.)

November 2022 — **Reproducible High Performance Computing Without Redundancy with Nix**, *IEEE PDGC 2022*, R. Goswami, Ruhila S. Goswami (née S.), A. Goswami, S. Goswami and D. Goswami

November 2022 — **Reproducible Literate Programming Workflows for Censored Data**, *IOP Machine Learning for Healthcare*, R. Goswami, R. S.

Honors and Awards

2025 — **Best Poster Award**, *Icelandic Ecological Society (Vistís)*

Research Experience

Summer 2023 — **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff
Studied the synteny of histone genes in salmonids, focusing on gene arrangement and neighboring genomic regions.

Winter 2022 — **Prof. Debabrata Goswami**, *IIT Kanpur*, Research Intern
Studied the basics of non-linear optics and applications of optical tweezers in cancer cell research and microscopy.

Winter 2022 — **Dr. Nagma Parveen**, *IIT Kanpur*, Research Intern
Studied wet-lab methods for viral mechanisms under external stimuli and began work on a SARS-CoV-2 pseudovirus protocol using an HIV-1 lentiviral packaging system with a luciferase reporter.

Summer 2022 — **Prof. Arnar Pálsson**, *University of Iceland*, Research Staff
Detailed analysis in a literate and reproducible manner for simulating a series of possible molecular evolutionary pathways for the *Salmonid* using phylogenetic trees. This involved the five steps on an HPC (High Performance Computer) with literate programming visualization in R:

- Data curation with NCBI databases
- Homology inference using similarity measures (BLAST)
- Multiple sequence alignment (MUSCLE5)
- Alignment trimming (Gblocks)
- Tree simulation with distance measures (BIONJ) and maximum likelihood approaches (RAxML-NG)

Project Report: Computational Primitives for Evolutionary Paths ($\approx$ 147 pages)

Summer 2021 — **Dr. Lolitika Mandal**, *IISER Mohali*, Research Intern
Explored genetic tools for working with *Drosophila* from a wet-lab perspective of data collection and analysis.

Teaching Experience

Spring 2026 — **LÍF401G Developmental Biology**, *University of Iceland*, Teaching Assistant
Course led by Prof. Adam Ray Smith. Assisted with embryo analysis for an 8 credit undergraduate developmental biology course.

Fall 2025 — **LÍF541G Plant Physiology**, *University of Iceland*, Teaching Assistant
Course led by Prof. Victor Anthony Albert. Led student discussions in a 7.5 credit undergraduate course focused primarily on plant genetics.

Spring 2025 **LÍF403G Evolutionary Biology**, *University of Iceland*, Teaching Assistant
8 credit undergraduate course on evolutionary theory, natural selection, adaptation, and the history of life.

Research Skills

Genomics: Population genetics, genomic synteny, structural variation, PCR-based sex determination, non-invasive sampling

Phenotypes: Dental and craniofacial variation, bone extraction, otolith extraction, fish dissection, embryo analysis

Evolutionary methods: Phylogenetic tree construction (distance, maximum likelihood, Bayesian), BLAST, MUSCLE5, Gblocks, RAxML-NG, BEAST2

Computing: R, Python, shell, Git, Snakemake, reproducible literate programming

Laboratory: DNA extraction and preservation, PCR, microscopy, *Drosophila* tissue dissection, cell culture

Field: Electrofishing, salmonid sampling and preservation

Service and Open Source

June-August 2024 **`pydSEAMSlib`: Enhanced Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
Improved usability and accessibility of pySEAMS through documentation, installation refinements, and user-facing features.

June-August 2023 **`PYSEAMS`: Python bindings for d-SEAMS**, *Google Summer of Code*, Student Developer
Developed Python bindings for d-SEAMS, a molecular dynamics analysis engine for qualitative ice structure classification in confined and bulk systems. Refactored C++ code, removed Lua bindings, improved macOS support, and contributed to documentation, automated testing, and repository maintenance.

2022–present **IEEE P3173**, *IEEE Standards Committee*, Secretary
Supporting the drafting of the IEEE Standard for Endocrine Disrupting Chemical Hazard Labeling.

Affiliations

2024–present **Nordic Society Oikos**, Student Member

2024–present **Icelandic Ecological Society (Vistís)**, Student Member

2024–present	**The Icelandic Biological Society (Líffræðigáttin)**, Student Member
2022–present	**British Ecological Society**, Student Member
2022–present	**Genetics Society, UK**, Undergraduate Member
2022–present	**Genetics Society of America**, Undergraduate Member

Certifications

April 2024	**Biophotonics**, *IIT Kharagpur*, Distinction, 87%
April 2024	**Introduction to Professional Scientific communication**, *IIT Kanpur*, 79%
Sep 2022	**Applications of machine learning techniques in biology using WEKA**, *IIT Madras*, Distinction, 93%
Nov 2022	**Practical Python for beginners: a biochemist's guide**, *Biochemical Society, U.K.*